

Project 8 - Strategy Evaluation

Edward Capone
ecapone3@gatech.edu

INTRODUCTION

This project integrates machine learning with investing theory by developing and comparing two trading strategies: a rule-based Manual Strategy and a Q-learning-based StrategyLearner. Both approaches evaluate performance over in-sample and out-of-sample periods using common technical indicators—Bollinger Bands, MACD, RSI, and Momentum. While the Manual Strategy uses predefined rules to generate trading signals, the StrategyLearner constructs a discrete state-space from these indicators for Q-learning to optimize trades. Our hypothesis was that the Manual Strategy would outperform the benchmark in-sample and that careful tuning of the StrategyLearner could further enhance returns. Although the learner achieved robust in-sample performance, its out-of-sample results revealed a significant gap, suggesting overfitting and challenges in generalization, as well as high sensitivity to transaction costs. This report details the design, implementation, and comparative evaluation of both strategies, including an overview of the selected indicators, a description of the manual rule-based system, a discussion on state-space discretization and hyperparameter tuning for the StrategyLearner, and analyses of experiments comparing strategy performance and impact sensitivity.

INDICATOR OVERVIEW

Bollinger Bands %B

Bollinger Bands combine a Simple Moving Average (SMA) with upper and lower bands set typically at two standard deviations, computed over an N-day window:

$$\text{SMA}(t) = \frac{1}{N} \sum_{i=t-N+1}^t P_i,$$

$$\text{Upper Band}(t) = \text{SMA}(t) + 2\sigma(t), \quad \text{Lower Band}(t) = \text{SMA}(t) - 2\sigma(t),$$

where $\sigma(t)$ represents the standard deviation of the price over the past N days. The Bollinger Band %B indicator normalizes the price relative to these bands:

$$\%B(t) = \frac{P(t) - \text{SMA}(t)}{2\sigma(t)}.$$

Adaptive thresholds based on the in-sample quantile distribution (typically the 20th and 80th percentiles) partition %B into low, medium, and high states. These

thresholds can be adjusted (e.g., to the 25th and 75th percentiles) to ensure a balanced state-space (StockCharts, 2024).

MACD

The Moving Average Convergence Divergence (MACD) is defined as the difference between a 12-day and a 26-day Exponential Moving Average (EMA):

$$\text{MACD}(t) = \text{EMA}_{12}(t) - \text{EMA}_{26}(t).$$

A positive MACD indicates bullish momentum, while a negative value suggests bearish conditions. In our implementation, we use a fixed threshold of zero to classify MACD into binary states, which are then integrated with other indicators (Investopedia, 2024).

Relative Strength Index (RSI)

RSI measures the speed and change of price movements:

$$\text{RSI}(t) = 100 - \frac{100}{1 + \text{RS}(t)},$$

where $\text{RS}(t)$ is the average gain to loss ratio over a 14-day period. Traditionally, RSI values above 70 indicate overbought conditions and below 30 indicate oversold conditions. We refine this by discretizing RSI into three levels using quantile-based thresholds (initially the 25th and 75th percentiles) from the in-sample data, thereby adapting the indicator to current market conditions (Investopedia, 2024).

Momentum

The Momentum indicator captures the rate of price change over a lookback period:

$$\text{Momentum}(t) = \frac{P(t)}{P(t - N)} - 1.$$

Positive momentum indicates strength in an upward trend, while negative momentum suggests weakness. We discretize momentum into three levels using quantile thresholds (initially set at the 30th and 70th percentiles) to normalize the indicator for robust integration into our trading state-space (Investopedia, 2023).

MANUAL STRATEGY

Overview and Trading Rules

Our manual strategy leverages four well-known technical indicators—Bollinger Bands %B, MACD, RSI, and Momentum—to generate trading signals without

using machine learning. Each indicator captures a key market aspect. Bollinger %B reflects volatility (with low values suggesting oversold conditions and high values overbought), MACD indicates trend direction, RSI gauges overbought/oversold states, and Momentum measures the speed of price changes. For each indicator, values are discretized into three levels using fixed or adaptive thresholds (e.g., Bollinger %B below the 20th percentile triggers a long signal, while above the 80th triggers a short signal). Each indicator effectively casts a "vote" on whether to buy, hold, or sell, and these votes are combined via a majority rule to determine the final trading decision—buying 1000 shares if the net signal exceeds a positive threshold, selling 1000 shares if it falls below a negative threshold, or holding otherwise. In our performance charts, vertical blue lines denote long entry points and black lines indicate short entry points.

Performance Comparison: In-Sample vs. Out-of-Sample

Figure 1 shows the normalized portfolio performance for the Manual Strategy and the benchmark during the in-sample period. As required, our rule-based manual strategy outperforms the benchmark in-sample. This is evidenced by the upward movement in the Manual Strategy's normalized values compared to the benchmark. However, when tested on out-of-sample data (Figure 2), the performance deteriorates; the Manual Strategy underperforms relative to in-sample results and falls short of the benchmark.

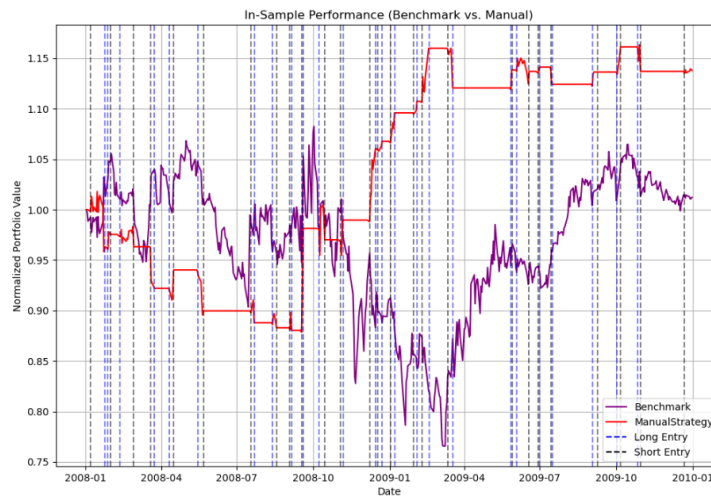


Figure 1—In-Sample Performance: Normalized Portfolio Values for Manual Strategy and Benchmark.

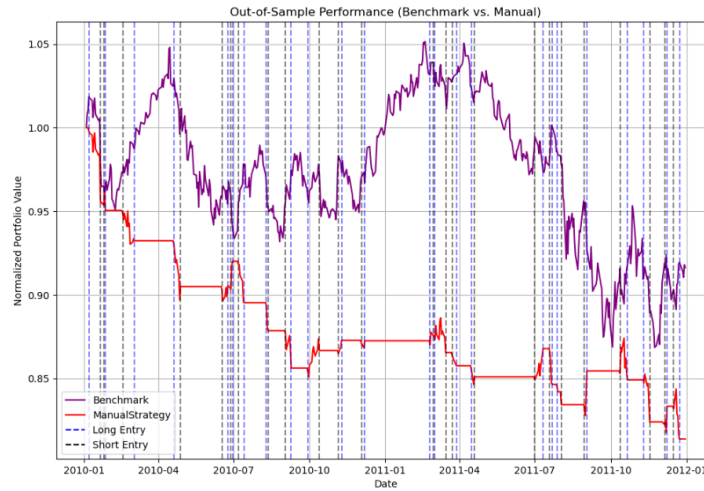


Figure 2—Out-of-Sample Performance: Normalized Portfolio Values for Manual Strategy and Benchmark.

Discussion and Analysis

The Manual Strategy’s design is rooted in established technical analysis principles. Its in-sample performance, which outperforms the benchmark, demonstrates that the selected indicators can provide actionable signals when combined appropriately. The strategy’s rules are based on threshold crossings and signal aggregation, which are straightforward to implement and understand, and is advantageous for transparency and reproducibility. However, the observed performance differences between in-sample and out-of-sample periods reveal some inherent limitations. In particular, the Manual Strategy’s out-of-sample underperformance relative to in-sample results suggests a lack of adaptability to shifting market conditions. Overfitting to the historical data is one plausible explanation: the thresholds and rules are optimized for the market environment experienced during the training period, but these conditions may not persist in future periods. Market regime changes, variations in volatility, and differences in trend strength can all contribute to the diminished performance seen out-of-sample. Moreover, the fixed nature of the rules might not capture nuanced market signals that evolve over time. The strategy uses static thresholds which, while effective in the in-sample period, may fail to adjust as market dynamics change. This underscores a key challenge in manual rule-based systems: the trade-off between simplicity and flexibility. While the

Manual Strategy is simple and interpretable, its inflexibility can be a disadvantage when facing dynamic market conditions.

STRATEGY LEARNER

Our Strategy Learner recasts trading as a Q-learning problem (Sutton and Barto, 2018), enabling the agent to learn profitable signals from historical data. We discretize four indicators—Bollinger Bands %B, MACD, RSI, and Momentum—into three levels (using adaptive quantiles for all except MACD, which is thresholded at zero), yielding $3^4 = 81$ combinations. By appending the current portfolio position (with three levels: $-1000, 0, +1000$), we form a unified state space of 243 states.

Q-Learning Hyperparameters and Tuning

Key hyperparameters include the learning rate (α), discount factor (γ), random action rate (rar) with decay (radr), and the number of dyna updates. We tuned these exclusively on in-sample data using a grid search, selecting parameters that provided strong cumulative returns and robust state coverage, which was verified by our Q-learner heatmap.

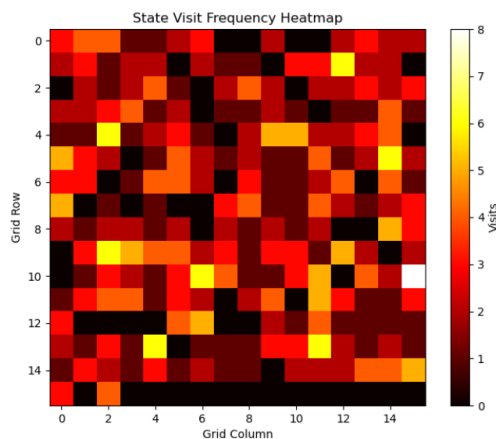


Figure 3—Q-Learner State Visit Frequency Heatmap.

This heatmap illustrates the distribution of visits across our Q-learning agent’s 243 discrete states. High-frequency states indicate market and position conditions the agent encounters repeatedly or explores thoroughly in pursuit of profitable trades. Observing such patterns helps identify potential overfitting (e.g., if states are rarely visited) and guides further tuning of indicators or learning parameters.

Weighted Lookahead Reward with Exponential Decay and Trade Penalty

The reward is computed over a 10-day horizon using exponential decay (with decay parameter $\lambda = 0.7$) to emphasize near-term returns:

$$\text{Reward} = \left(\frac{\sum_{j=1}^H \exp(-\lambda(j-1)) \Delta P_j}{\sum_{j=1}^H \exp(-\lambda(j-1))} \right) \times \text{current position} - \text{Cost}.$$

A trade penalty (tuned between 0.0010 and 0.0015) is applied to discourage marginal trades and control transaction costs.

Challenges and Design Rationale

Balancing in-sample performance with generalization is challenging. While our configuration yields robust in-sample returns, the out-of-sample gap suggests overfitting—a common issue in Q-learning for trading, given the non-stationary nature of financial markets. Our adaptive thresholds and exponential decay reward function help capture persistent trends and reduce short-term noise. However, the discretization of continuous indicators into a finite state space may oversimplify complex market dynamics, and the model remains sensitive to transaction costs and variations in market impact. Despite these challenges, our design meets project requirements by tuning on in-sample data and rigorously comparing performance across periods, thereby highlighting both the potential and limitations of applying Q-learning to trading.

EXPERIMENT 1: MANUAL STRATEGY VERSUS STRATEGYLEARNER

In this experiment, we compare our Manual Strategy and StrategyLearner for JPM using a common set of technical indicators. The data is divided into an in-sample period (January 1, 2008 – December 31, 2009) for training and tuning, and an out-of-sample period (January 1, 2010 – December 31, 2011) for evaluation. Our hypothesis is that a rule-based Manual Strategy, which combines signals from Bollinger Bands %B, MACD, RSI, and Momentum, will outperform a buy-and-hold benchmark in-sample. Meanwhile, the Q-learning-based StrategyLearner, which creates a state space of 243 states using adaptive quantile thresholds and employs a reward function based on a 10-day weighted lookahead, is expected to capture more subtle market patterns.

The candidate hyperparameters include:

- **In-Sample Period:** January 1, 2008 – December 31, 2009.

- **Out-of-Sample Period:** January 1, 2010 – December 31, 2011.
- **Indicators:** Bollinger Bands (20th/80th percentiles), RSI (25th/75th percentiles), and Momentum (30th/70th percentiles).
- **Reward Function:** 10-day weighted lookahead with exponential decay (decay parameter = 0.7).
- **Other:** Commission \$9.95, Impact 0.005, Minimum Holding Period 5 days.

Figure 4 shows the normalized portfolio values (with an initial value of 1.0) for the benchmark, Manual Strategy (red), and the StrategyLearner (green) for the in-sample period. As observed, the Manual Strategy outperforms the benchmark in-sample, and the StrategyLearner achieves even higher in-sample cumulative returns (approximately 13% in our latest run).

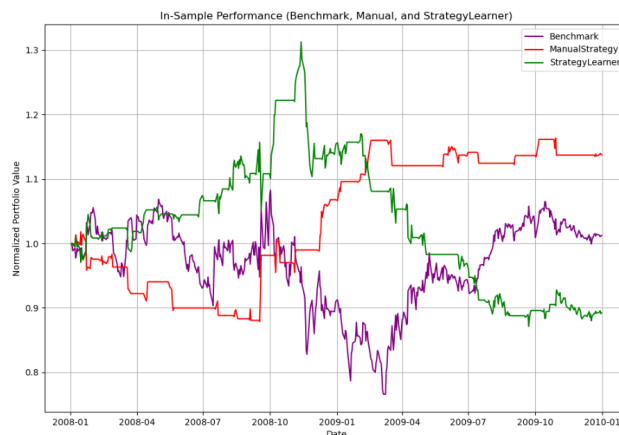


Figure 4—Normalized In-Sample Performance for Manual Strategy, StrategyLearner, and Benchmark.

Figure 5 illustrates the out-of-sample performance. While the Manual Strategy and benchmark perform negatively, the StrategyLearner also exhibits negative returns out-of-sample, indicating an overfitting issue. These differences may occur because the patterns learned during the in-sample period do not persist in the out-of-sample period due to changing market conditions. Variability in market regimes and shifts in volatility can cause the previously optimized rules to underperform in new environments.

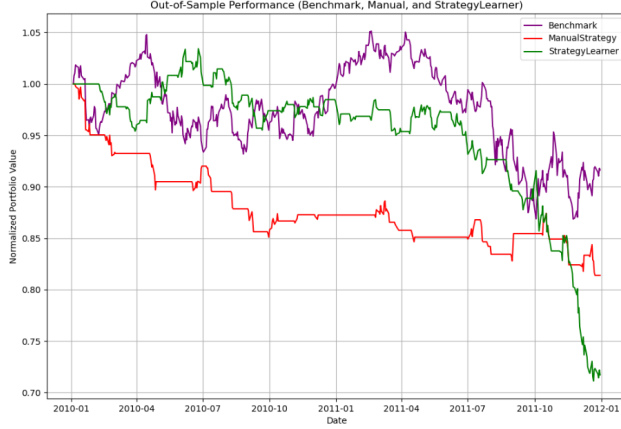


Figure 5—Normalized Out-of-Sample Performance for Manual Strategy, StrategyLearner, and Benchmark.

EXPERIMENT 2: IMPACT SENSITIVITY FOR STRATEGYLEARNER

This experiment investigates how varying the market impact parameter affects the in-sample performance of the StrategyLearner. Our hypothesis is that as the impact cost increases, the trading behavior and cumulative returns of the StrategyLearner will change significantly. We evaluate at least two key metrics: **cumulative return** and **trade count**. For each metric, we consider measurements at several different impact levels. The chosen impact values are 0.0, 0.005, 0.01, and 0.02. For this experiment, the StrategyLearner is trained solely on in-sample data (January 1, 2008 to December 31, 2009) with a fixed set of other hyperparameters. We then test the policy on the same in-sample period for consistency. In our experiments, several parameters were fixed: we set the commission at \$9.95, enforced a minimum holding period of 5 days, and selected adaptive thresholds for our indicators. The reward function is defined as a weighted lookahead with exponential decay over a 10-day horizon. In contrast, the market impact is treated as a variable parameter with values of 0.0, 0.005, 0.01, and 0.02. For each impact level, we recorded the cumulative return (computed as the relative change between the final and initial portfolio values), the trade count (i.e., the total number of days on which a nonzero trade occurred), and the standard deviation of daily returns as a measure of the strategy’s volatility. Table 1 summarizes our results for each impact level. As expected, the StrategyLearner’s performance is highly sensitive to transaction costs. With no impact cost, the learner tends to produce a relatively high cumu-

lative return due to more aggressive trading. However, as the impact increases, the cumulative return declines, and the strategy becomes less profitable or even unprofitable.

Impact	Cumulative Return	Trade Count
0.000	51.83%	63
0.005	-10.69%	59
0.010	-20.65%	51
0.020	-30.51%	52

Table 1—Experiment 2 Summary: Sensitivity of StrategyLearner to Impact.

Figure 6 illustrates how cumulative return and trade count vary with market impact. The results show that even a small impact value of 0.005 results in a dramatic drop in returns—from a strong gain at zero impact to a significant loss—while also affecting the number of trades executed. This confirms our hypothesis that the StrategyLearner is highly sensitive to market impact, underscoring the importance of carefully calibrating the reward function and trade penalties.

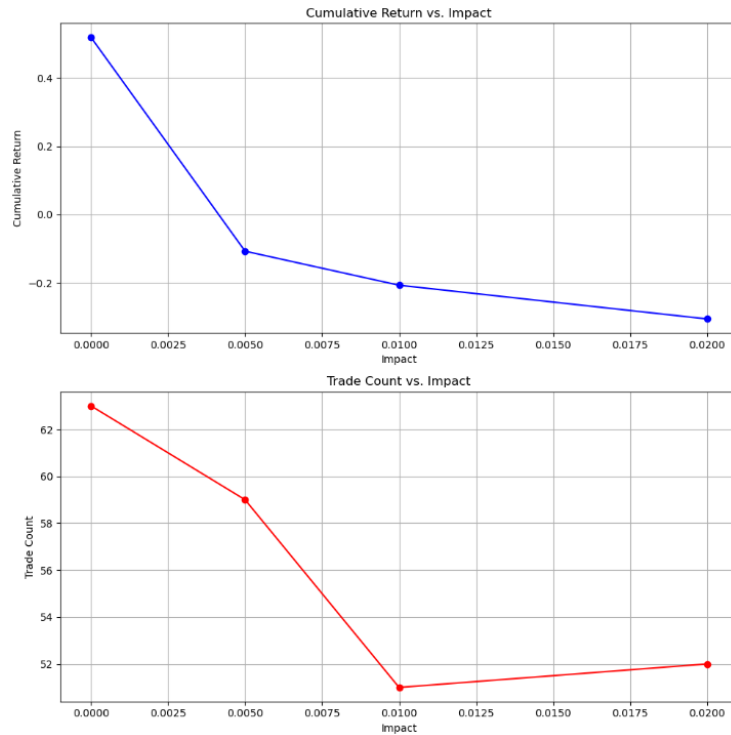


Figure 6—Cumulative Return vs. Impact.

In conclusion, Experiment 2 illustrates that the performance of the StrategyLearner is highly dependent on transaction cost parameters. These findings underscore the challenge of developing a policy that is robust to varying market conditions. An informed reader could replicate this experiment by training the StrategyLearner with the fixed hyperparameters and then varying the impact parameter as described, subsequently recording the cumulative return, trade count, and volatility metrics.

CONCLUSION

In this report, we developed and evaluated two distinct trading strategies—a rule-based Manual Strategy and a Q-learning-based StrategyLearner. We did this in an effort to investigate their performance against a benchmark over both in-sample and out-of-sample periods. Our Manual Strategy, which combines signals from Bollinger Bands %B, MACD, RSI, and Momentum using adaptive thresholds and majority-vote rules, demonstrated clear in-sample outperformance relative to a buy-and-hold approach. However, its performance deteriorated out-of-sample, highlighting the challenges associated with overfitting and adapting to dynamic market conditions. Conversely, the StrategyLearner, formulated through a discretized state space and optimized via Q-learning, achieved competitive in-sample results. Yet, its sensitivity to market impact, which was exhibited through a detailed sensitivity analysis, revealed that even minor variations in impact parameters can significantly influence cumulative returns and trading frequency. This finding underlines the critical need to incorporate transaction cost considerations effectively in automated trading systems. Overall, our comparative study illustrates that while both strategies have merits, the rule-based Manual Strategy is more transparent and easier to interpret, whereas the StrategyLearner offers potential for higher returns under controlled conditions but faces considerable challenges in generalizing to out-of-sample data. These insights reinforce the importance of balancing model complexity, robustness, and adaptability when integrating machine learning with trading. Future work may focus on enhancing model generalization and fine-tuning hyperparameters to better manage market impact and reduce overfitting.

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